

Basic LAMMPS & NVE Simulation of LJ System

Tutors:

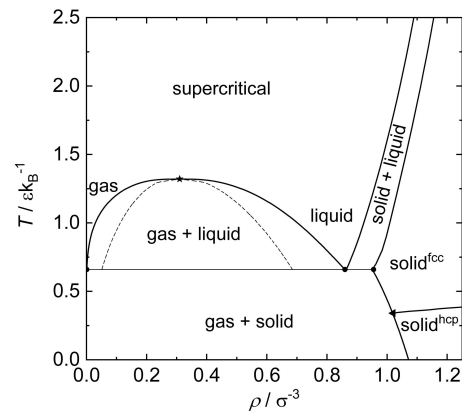
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Content

1. Input File
2. Run NVE Simulation
3. Output
4. Jupyter Notebook
5. Sample Phase Diagram of LJ system (ρ , T)



Input File

- Order Matters!

# 1) Initialization	→	Global Parameters
# 2) System definition	→	Simulation Box and Position of Atoms
# 3) Settings	→	Atoms Parameters
# 4) Monitoring	→	Outputs
# 5) Run	→	Type of Simulation

Input File (in.nve)

```
# 1) Initialization  
units lj  
dimension 3  
atom_style atomic  
boundary p p p
```

Input File (in.nve)

```
# 2) System definition
variable rho    equal ???    # density
lattice fcc ${rho}
region simbox block 0 6 0 6 0 6
create_box 1 simbox
create_atoms 1 box
```

Input File (in.nve)

```
# 3) Settings
variable T0      equal ???  # initial temperature
mass 1 1.0
pair_style lj/cut 4.0
pair_coeff 1 1 1.0 1.0
velocity all create ${T0} 12345 mom yes rot yes dist gaussian
```

Input File (in.nve)

```
# 4) Monitoring
variable target_phase string ??? # suffix in the data

thermo 100
thermo_style custom step temp etotal pe ke

dump wrap    all custom 10 wrap_pos_${target_phase}.lammprj id x y z
dump_modify wrap    sort id
dump unwrap  all custom 10 unwrap_pos_${target_phase}.lammprj id xu yu zu
dump_modify unwrap sort id
dump vels    all custom 10 vels_${target_phase}.lammprj id vx vy vz
dump_modify vels    sort id

variable mytemp    equal temp
variable myetot    equal etotal
variable mype      equal pe
variable myke      equal ke
fix thermofile all ave/time 10 1 10 &
    v_mytemp v_myetot v_mype v_myke &
    file thermo_${target_phase}.dat
```

Input File (in.nve)

```
# 5) Run  
fix mynve all nve  
timestep 0.005  
run 20000
```


Run Simulation

```
lmp -in in.nve
```

Output

Jupyter Notebook

Sample Phase Diagram

1. Choose a target phase
2. Modify the input file (ρ , T_0 , suffix)
3. `lmp -in in.nve`
4. Separate equilibration from production
5. Does $\langle T \rangle_{\text{prod}}$ correspond to your target phase?

